

Student Worksheet: Radioactive Decay & Mystery Source Identification

Name: _____ Date: _____ Class: _____

1 Objective

The objective of this experiment is to:

1. Investigate the penetration power and shielding requirements for Alpha (α), Beta (β), and Gamma (γ) radiation.
2. Experimentally determine the half-life ($t_{1/2}$) of a randomly assigned Mystery Source.
3. Identify the nature of the Mystery Source based on experimental shielding tests.

2 Part A: Reference Testing & Shielding Observations

Before testing the mystery source, calibrate the Geiger-Müller counter and test the three known sources. Record whether the radiation is **Stopped**, **Reduced**, or **Unaffected** by the shield.

Radiation Type	Paper Shield	Aluminum Shield	Lead Shield
Alpha (α)	_____	_____	_____
Beta (β)	_____	_____	_____
Gamma (γ)	_____	_____	_____

3 Part B: Mystery Source Data Acquisition

Select a Mystery Source from the simulation and record your specific observations below.

Mystery Source ID/Number (if applicable): _____

3.1 Shielding Analysis

Place each shield in front of your Mystery Source and record the impact on the count rate.

- ☐ **Count Rate (No Shield):** _____ counts/sec
- ☐ **Count Rate (Paper):** _____ counts/sec
- ☐ **Count Rate (Aluminum):** _____ counts/sec
- ☐ **Count Rate (Lead):** _____ counts/sec

3.2 Half-Life Determination

Using the counts-vs-time graph generated by the simulation, determine the time required for the activity to reach 50% of its initial value.

- ☐ **Initial Activity (N_0):** _____ counts/sec
- ☐ **Target Activity ($0.5 \times N_0$):** _____ counts/sec
- ☐ **Measured Half-Life ($t_{1/2}$):** _____ seconds

4 Calculations & Analysis

Show your work for the following calculations based on your specific Mystery Source data.

1. **Calculate the Decay Constant (λ):**

Use the formula: $\lambda = \frac{\ln(2)}{t_{1/2}}$

$$\lambda = \underline{\hspace{2cm}}$$

2. **Predicting Future Activity:**

Using the decay law $N(t) = N_0 e^{-\lambda t}$, calculate what the activity of your source will be after **three** half-lives have passed.

$$N(3t_{1/2}) = \underline{\hspace{2cm}}$$

5 Conclusion & Identification

1. **Nature of Source:** Based on your shielding tests, what type of radiation does your Mystery Source emit?

2. **Justification:** Explain how the shielding data supports your conclusion.

3. **Summary of Half-Life:** State your final measured half-life and compare it to the expected range (10 – 30s).
